Assessment 1 – Part 1

Overview

In this assignment you will write formulas in first-order logic that represent statements made in English. You will prepare your answers in the format that Prover9 uses to express first order logic. There is only one problem in this part of the assignment and you must submit via Gradescope one file that follows exactly the format explained below. Your submission will be automatically marked using Prover9. The automatic marking will accept any formula that is logically equivalent to an ideal solution. Representation can be done in many different ways and the marks are for the logical idea behind the statements not for a specific way of formulating something.

The problem has this structure:

- A number of statements in English (and some mathematical notation) that you have to represent as assumptions for Prover9
- One further statement that you formulate as the goal of an argument.
- The language to be used in the representation is specified for you. You can only use the predicate symbols that are given and no others. You can use all the logical connectives and quantifiers in first order logic.
- There is a template file that you must edit to make your submission. The file explains where you input your answer.

Because the submissions are to be marked by running Prover9 on them you should make use of Prover9 when preparing your answers to avoid syntax errors and to check that the goal follows from the assumptions. The template file (when you have inserted all your answers) can be input directly into Prover9 without any further changes.

Your submission will be exactly one file that must have precisely the following name. Submissions with the wrong file name will cause the autograder to award zero marks.

assessment1_part1.p9

Problem: Modelling propositional logic

Outline

In this problem you can think of the individuals in the domain of first order logic as propositional statements.
There is a unary relation symbol True and the formula True(x) means that the statement x is true.
There is a ternary relation symbol Conjunction and the formula Conjunction(x, y, z) means that x is a conjunction of statements y and z. Because of the way we are using Conjunction as a relation, we need axioms A1 and A2 below to ensure we can think of Conjunction(x, y, z)as modelling the idea $x = y \wedge z$.
There are also relation symbols Disjunction and Negation to model disjunctions and negations. The ultimate goal is to show that $\neg(y \wedge z) \equiv \neg y \vee \neg z$.

The statements to be represented

The assumptions

A1: Any two statements always have a conjunction.

A2: Two statements, in a given order, have at most one conjunction.

A3: A conjunction of two statements is true if and only if both the individual statements are themselves true.

A4: Any two statements always have a disjunction.

A5: Two statements, in a given order, have at most one disjunction.

A6: A disjunction of two statements is true if and only if at least one of the individual statements is itself true.

A7: Any statement always has a unique negation.

A8: A negation of a statement is true if and only if the statement itself is false.

A9: The formula `NegatedConjunction(x,y,z)` models $x = \neg(y \wedge z)$.

A10: The formula `DisjunctiveNegations(x,y,z)` models $x = \neg y \vee \neg z$.

The goal

G: The statements $\neg(y \wedge z)$ and $\neg y \vee \neg z$ always have the same truth value.

The Language you must use

No other relation symbols at all, and the exact spelling below really matters

Property Predicate (Unary Relation Symbol): True

Relation Predicates (Binary Relation Symbol): Negation

Equality Predicate (special Binary Relation Symbol): =

Relation Predicates (Ternary Relation Symbols):

Conjunction, Disjunction,

`NegatedConjunction, DisjunctiveNegations`